

Each question is worth 5 points for a total of 30 points. The extra Credit can get you an additional 5 points making the highest possible score 35 out of 30. Write solutions using COMPLETE sentences.

Problem 1

Let (S, P) be the sample space in which $S = \{1, 2, 3, 4\}$. Suppose $P(1) = x$, $P(2) = 2x$, $P(3) = 3x$, and $P(4) = 4x$. Find x .

Problem 2

A coin is flipped four times. Let A be the event that we record an equal number of *HEADS* and *TAILS*.

a) Write down the event A by explicitly writing its elements between curly braces.

b) Evaluate $P(A)$.

Problem 3

A pair of fair dice are rolled.

- a) Write down the event space (S, P) .
- b) Write down the event A , that the sum of the dice is 7.
- c) Write down the event B , that neither die shows a 2.
- d) Evaluate $P(A)$
- e) Evaluate $P(B)$.
- f) What is the probability that they sum to 7 given that neither die shows a 2?

Problem 4

A fair coin is flipped three times. This is modeled by a sample space (S, P) where S contains the eight lists from HHH to TTT , all with probability $\frac{1}{8}$. Let X denote the number of times we see *TAILS*.

- a) Write X explicitly as a function defined on S .
- b) Write the event X is odd as a set.
- c) Calculate $P(X \text{ is odd})$.

Problem 5

You play a game in which you roll a die and you win (in dollars) the square of the number on the die. For example, if you roll a 5, then you win \$25. On average, how much money would you expect to receive per play of this game?

Problem 6

Let (S, P) be a sample space where $S = \{1, 2, 3, \dots, 10\}$ and $P(j) = \frac{j}{55}$ for $1 \leq j \leq 10$. Let A be the event $A = \{1, 4, 7, 9\}$ and let B be the event $B = \{1, 2, 3, 4, 5\}$.

a) What is the probability of A ?

b) What is the probability of B ?

c) What is $P(A|B)$?

d) What is $P(B|A)$?

Extra Credit:

A coin is flipped ten times. Let X_1 be the number of times we see *HEADS* immediately before *TAILS* and let X_2 be the number of times we see *TAILS* immediately before we see *HEADS*. For example, if we flip *THHTTHHTHH*, then $X_1 = 2$ and $X_2 = 3$ because we have *HT* twice and *TH* three times in *THHTTHHTHH*. Are X_1 and X_2 independent random variables? Prove your answer.